

TARANDEEP SINGH JUNEJA

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Summary

Final-year B.Tech Computer Science student at Vellore Institute of Technology, Bhopal, seeking challenging roles in software development. Possess strong analytical and problem-solving skills, demonstrated through 500+ DSA problems. Gained practical experience at IIT Bombay (Osdag project) and as a Freelance Prompt Engineer at Outlier AI, with hands-on expertise in Python, SQL, Cloud Computing, Machine Learning, Deep Learning, and full-stack development.

Work Experience

FOSSEE Summer Intern

Feb 2025 – Jun 2025

IIT Bombay

- Developed Plate Girder GUI module following established design patterns, extending Osdag's structural design capabilities.
- Implemented bolted butt joint calculations with geometric validation and IS 800:2007 code compliance.
- Enhanced Tension Member module by adding flat plate option with freeze/unfreeze functionality for parameter locking.

Software Taxonomy Analyst (Freelancer)

Sep 2024 – Jan 2025

Outlier AI – Code Classification

Remote

- Classified and rated 100+ code snippets across C++, Python, and JavaScript using a structured taxonomy (L0–L3) for AI training datasets.
- Documented reasoning behind ratings (e.g., why a snippet was rated 1 vs. 5) to improve model consistency and code understanding.

Projects

CNN Audio Classification Visualizer — Python, PyTorch, FastAPI, Modal — [GitHub]

Sep 2024

- Implemented ResNet-based CNN model achieving **83.5% accuracy** on ESC-50 dataset (50 sound classes).
- Built audio preprocessing pipeline with Mel-spectrograms and OneCycleLR scheduling.
- Implemented real-time CNN feature visualization for deep learning explainability.

Credit Card Fraud Detection System — Python, Scikit-learn, Streamlit — [GitHub] — [Live Demo]

Nov 2024

- Developed ML-based fraud detection system on **284,807 transactions** with 92% accuracy using Logistic Regression.
- Solved class imbalance (0.17% fraud rate) with undersampling for balanced training.
- Deployed interactive Streamlit application for real-time fraud prediction using PCA-transformed features.

Omdena Road Safety Analytics model — Python, Roboflow, Pandas, NumPy, Scikit-learn — [GitHub] Jun 2024 – Sep 2024

- Collaborated with a 50+ member global team to enhance road safety using AI-driven accident analysis.
- Labeled and trained a dataset of **5,000+ road accident images** on Roboflow to improve model accuracy.

Technical Skills

Languages: C++, Python, SQL

Frameworks/Tools: Streamlit, FastAPI, PyTorch, Scikit-learn, Git, GitHub, Postman, Modal (serverless)

Core Areas: Backend, Frontend, Problem-Solving, Machine Learning, Deep Learning, Open-source

Education

Vellore Institute of Technology, Bhopal

Aug 2022 – Jul 2026

B.Tech in Computer Science and Engineering

CGPA: 8.53/10.0

- **Relevant Coursework:** Data Structures and Algorithms (DSA), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks (CN), Object-Oriented Programming (OOP)

Certifications & Achievements

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **Aug 2025**
- Oracle Autonomous Database Cloud 2025 Certified Professional **Aug 2025**
- Postman API Fundamentals Student Expert — Postman Academy **Jul 2024**
- Adobe GenSolve Hackathon Semi-Finalist — Top finalist among 5000+ participants **2024**
- Infosys HackWithInfy 2025 – Pre-Placement Opportunity — Secured a pre-placement offer for the Specialist Programmer role by clearing Round 1 of HackWithInfy, a national coding competition with 200,000+ participants **2025**